

Docket No. SA-537

Exhibit No. 13-B

NATIONAL TRANSPORTATION SAFETY BOARD

Washington, D.C.

Aircraft Performance Group Study

(23 Pages)

December 6, 2013

Aircraft Performance Group Study

I. ACCIDENT

Description:	Impact with Sea Wall during Final Approach to Runway 28L
Location:	San Francisco International Airport (KSFO), San Francisco, CA
Date:	July 6, 2013
Time:	1128 Pacific Daylight Time (PDT)
Aircraft:	Boeing 777-200ER, HL7742
Operator:	Asiana Airlines
NTSB Number:	DCA13MA120

II. AIRCRAFT PERFORMANCE GROUP

Chairman:	Kevin J. Renze, Ph.D. Vehicle Performance Division, RE-60 National Transportation Safety Board (NTSB)
Members	Robert Stoney Federal Aviation Administration (FAA) Test Pilot Seattle Aircraft Certification Office (ACO) Dae Young Lee Investigator Aviation and Railway Accident Investigation Board (ARAIB) Republic of Korea Sang Yoon Lee Flight Operations Quality Assurance (FOQA) Manager Asiana Airlines Paul J. Bolds-Moorehead 777 and 787 Model Lead, Stability & Control Aerodynamics Fleet Support The Boeing Company

1.0 INTRODUCTION

On July 6, 2013 at 11:28 am Pacific daylight time, a Boeing 777-200ER, registration HL7742, operated by Asiana Airlines as flight 214, struck the seawall short of runway 28L at San Francisco International Airport. The airplane was destroyed by impact forces and fire. Three of the 291 passengers were fatally injured. The flight was a regularly scheduled passenger flight from Incheon International Airport (RKSI), Seoul, Korea, and was operated under the provisions of 14 *Code of Federal Regulations* Part 129. Visual meteorological conditions prevailed at the time of the accident.

The event flight time alignment between the Cockpit Voice Recorder (CVR), Flight Data Recorder (FDR), Air Traffic Control (ATC) communication recordings, ATC Airport Surveillance Radar (ASR), and ATC Automatic Dependent Surveillance-Broadcast (ADS-B) data was determined. The flight 214 weight and balance records and component weight buildup was examined. A comparison of the maximum landing gear and flaps extended

speed placards to the documented flight 214 landing gear and flap configurations indicates no landing gear or flap extension placard exceedances. The B777-200ER low speed amber band parameter time history was reconstructed and compared to available FDR data for minimum maneuvering speed during the flight 214 approach and CVR data for an EICAS caution alert event and a stick shaker activation event. Finally, calculated pilot eye height, calculated Precision Approach Path Indicator (PAPI) guidance, and select FDR parameters were plotted to visualize the airplane flight path and energy state as a function of Asiana flight 214 distance from the runway 28L displaced threshold and time prior to impact.

The NTSB and the Boeing calculated minimum maneuver speed profiles compare favorably (within about 1 to 4 knots) with the sparse Flight Management Computer (FMC) minimum maneuver speed data recorded on the FDR. In addition, the calculated timing results for the expected Engine Indicating and Crew Alerting System (EICAS) caution alert AIRSPEED LOW and the stick shaker activation events appear to match the available CVR data within expected timing tolerances for each event.

2.0 TECHNICAL METHODS

The data sources and technical methods used in this study are described in this section.

2.1 Time Alignment

The CVR, FDR, ASR radar, ADS-B, ATC communications recordings, ACARS, and video surveillance camera data are each recorded as a function of time. However, the “timestamp” value associated with these data is not necessarily explicit. Moreover, data from two or more independent sources that record explicit timestamp information are generally not perfectly synchronized to a common or desired reference time basis. In particular, the CVR and the FDR do not explicitly record a time value with each respective event or data sample. In contrast, each record in a radar data log, ATC transcript, many security video frames, and ACARS entry is associated with an explicit time value, but the time reference used for each may not be synchronized to a common reference time basis.

The time alignment process yields one or more equations that correlate recorded information from independent sources to a reference clock, thereby defining the accident event sequence and elapsed time between events. The reference clock selected is generally the ATC radar or ADS-B data time. In this study the reference clock was based on the ADS-B data time.

The ability to correlate information between any two independent sources is dependent on the existence of at least one common event between the two sources. In practice, the FDR discrete parameters that indicate the activation of the Captain’s or First Officer’s VHF radio transmit keys provide common event sources because the begin- and end-keying events are typically recorded on the CVR and the FDR. The communication that occurs between the begin- and end-keying event is typically recorded on both the CVR and the ATC recordings.

2.1.1 Aircraft Recorder Data

The CVR transcript data and the FDR data for the accident flight are documented in the Cockpit Voice Recorder Factual Report and Transcript and the Flight Data Recorder Group Chairman’s Factual Report, respectively.

2.1.2 CVR and FDR Alignment

Common events on the CVR and FDR were identified and used to define the CVR/FDR time alignment constraints. Each FDR common event has an associated begin and end time value due to the discrete nature of the FDR data. Corresponding CVR common events are associated with a CVR time value which must be precisely measured. The points on a common event plot of FDR time versus CVR time define the constraints. A curve that satisfies all the constraints defines a mapping from CVR time to FDR time.

2.1.3 FDR and ATC Alignment

The existence of one or more common events between the ATC transcript and the CVR and the CVR/FDR time alignment permits the FDR data to be aligned to the ATC transcript. Time alignment between the ATC transcript and the FDR data is accomplished by choosing a specific ATC clock time to equal a specific FDR time.

2.1.4 FDR and Radar/ADS-B Data Alignment

The FDR and ADS-B data were time aligned using the FDR pressure altitude and the ADS-B altitude information. Given local barometric pressure and knowledge of the pressure reference used on the aircraft, the FDR pressure altitude was corrected for atmospheric deviation from standard day conditions. The ADS-B and FDR time alignment was then accomplished by 1) creating an overlay plot of the pressure altitude versus time from each source, 2) adding a +/-50 foot error band to the ADS-B altitude data, and 3) time shifting the ADS-B data with respect to the corrected FDR altitude data to a point where the FDR altitude curve was bounded by the +/-50 foot error band throughout the time period of interest.

2.1.5 FDR and Security Video Alignment

Security camera video recorded at the KFSO Security Operations Center and other video sources documented in the Attachments 11-21 of the Aircraft Performance Group Crash Site Factual Report have not been explicitly time aligned to date.

2.2 Airplane Weight and Balance

The accident aircraft weight was estimated by itemizing the aircraft component weights. Component weights include the aircraft empty weight, the crew weights including nominal crew baggage, passenger weights including carry-on baggage and personal effects, cargo, and fuel. The component weight buildup approach sums the component weights to arrive at a weight estimate for the event condition, accounting for weight changes from a reference condition. The HL7742 reference condition is the aircraft weight just prior to flight 214 engine start. Given the component weights and locations, the aircraft moment balance can be calculated.

The longitudinal center of gravity position in percent Mean Aerodynamic Chord (%MAC) and Index Units (I.U.) can be calculated using the following equations, given the moment arm in inches and the weight (for the latter case):

$$\%MAC = \{ [ARM \text{ (Inch)} - 1174.5] / 278.5 \} \times 100$$

$$\text{Index Units} = \{ WT \text{ (lb)} \times [ARM \text{ (Inch)} - 1258] \} / 400,000 + 50$$

2.2.1 Operating Empty Weight

According to Asiana Airlines weight and balance record AAR-QA-529-02, the HL7742 basic empty weight on September 6, 2007 was 309,752.47 pounds with the center of gravity located at 1232.42 inches aft of the reference datum. Asiana Airlines computes individual aircraft weights in accordance with their weight and balance control program. The flight 214 load sheet reflects a specific aircraft operating empty weight of 325,008 pounds, which includes crew weights for 2 flight crewmembers, 12 flight attendants, and standard operating items.

2.2.2 Crew and Crew Baggage Weights

Asiana Airlines uses crew weights (including bags) of 210 pounds per flight crewmember and 165 pounds per flight attendant, consistent with the guidance published in their weight and balance control program.

2.2.3 Passenger Weights

The Asiana Airlines average weight method for international flights accounts for nominal weights of 75 kg (165.3 pounds) per adult, 37 kg (81.6 pounds) per child (between 2 years and less than 13 years of age), and 10 kg (22 pounds) per infant (less than 24 months old). The average weight method includes a 22 pound-per-passenger allowance for carry-on baggage and personal effects.

The flight 214 load sheet reflects a total of 291 passengers (260 adults, 30 children, and 1 infant) and 2 reserve flight crewmembers on board. The total weight of 45,790 pounds for 293 persons is based on 262 adults ($262 \times 75 = 19,650$ kg), 30 children ($30 \times 37 = 1,110$ kg), and 1 infant ($1 \times 10 = 10$ kg) which yields 20,770 kg (45,790 pounds).

2.2.4 Fuel Weight

According to the flight 214 load sheet, the planned takeoff fuel for flight 214 was 193,200 pounds. Asiana Airlines assumes a 1,000 pound nominal taxi fuel burn for the Boeing 777, yielding a ramp weight of 194,200 pounds.

Prior to flight 214, HL7742 completed flight OZ111 from Kansai International Airport (RJBB) to Incheon and arrived with 14,100 pounds of fuel remaining based on ACARS data. A fuel receipt record documented in the Aircraft Performance Group Accident Site Factual Report shows that a total of 27,685 gallons of fuel (density 6.58 pounds per gallon) were uploaded in Incheon prior to the flight 214 departure. Based on the reported fuel density, HL7742 would have had 196,267 pounds of fuel (remaining fuel plus fuel added is $14,100 + 27685 \times 6.58$) available at flight 214 engine start for nominal conditions. The FDR data at engine start indicate 62,200, 62,500, and 69,700 pounds of fuel in the left wing tank, right wing tank, and center tank, respectively, for a total ramp fuel load of 194,400 pounds.

During the post-accident airplane recovery effort, the fuelers reported that the center fuel tank was empty, the number 1 fuel tank indicated between 5,000 and 6,000 pounds, and the number 2 fuel tank showed about 8,000 pounds. The fuelers noted that the airplane was not in a wings level attitude when these fuel values were measured.

2.2.5 Cargo Weight

Asiana Airlines accounts for actual weights for cargo and checked baggage. For flight 214 the load sheet documents 304 checked bags that weighed 10,245 pounds (33.7 pounds per bag average) and 12,461 pounds of freight. The total cargo compartment load of 22,706 pounds (baggage and freight) was distributed as follows: 3,219 pounds in compartment 1; 7,040 pounds in compartment 2; 1,903 pounds in compartment 3; 10,435 pounds in compartment 4; and 111 pounds in compartment 5.

2.2.6 Boeing Estimate of Airplane Weight and Center of Gravity Position during the Flight

At NTSB request Boeing Engineering estimated the flight 214 weight and balance based on the following fuel burn assumptions during the flight:

- a) Burn fuel from the center tanks until the center tanks are empty. In practice, some fuel remains in the center tanks at the time that fuel is switched to burn from the main tanks. However, the remaining center tank fuel is scavenged to the main tanks while burning from the mains. Fuel remaining in the center tanks when the tanks are switched has no effect on the time of arrival center of gravity calculation because that fuel would have been completely scavenged from the center tanks by the time the airplane lands.
- b) Burn fuel from the main tanks for the remainder of the flight.
- c) Center of gravity calculations equated to a flaps up configuration with landing gear retracted. Extending flaps and landing gear has a negligible net effect because the combined flap and landing gear center of gravity change is small. It is on the order of 0.5 %MAC in the aft direction.
- d) Fuel density is 6.8 pounds/gallon.

2.3 Speed Tape Parameter Calculations (Low Speed Amber Band)

Boeing provided the NTSB with the technical specifications required to reconstruct the speed tape time history of low speed amber band information expected to be presented to the flight crewmembers during the approach. The top of amber band speed is set to the Maneuver Margin above Stick Shaker (V_{SSNG}) value, which can be equated to a normal load factor maneuvering capability of 1.3 g. The minimum speed limit (V_{MIN}) is set to the computed Stick Shaker Speed (V_{SS}) value, which corresponds to the junction between the bottom of the amber band and the top of the red dashed line where stick shaker is scheduled to activate. Boeing and NTSB engineers independently implemented the relevant speed tape parameter technical specifications¹ using the available Asiana flight 214 FDR data.

For the 777-200ER, the calculated stick shaker speed is a function of the following variables: angle of attack, Mach number, calibrated airspeed, slat position, flap position, landing gear position, anti-ice configuration, speedbrake position, and engine thrust. Calculated stick shaker speed values are filtered prior to presentation to the flight crewmembers on the speed tape. For calculation purposes, if the slats are in-between detents, they are assumed to be in the lesser (less extended) detent. In contrast, if the flaps are in-between detents, they are assumed to be in the higher (more extended) detent.

¹ The B777 speed tape technical specifications are Boeing proprietary data and are not available for public release.

Stick shaker is activated if the angle of attack equals or exceeds the calculated stick shaker trip point angle of attack (AOA_{SS_TRIP}) and the airplane Air Mode Logic yields a value of Airplane In Air. The computed value of unfiltered stick shaker speed ($V_{SS_UNFILTERED}$) is latched if the angle of attack equals or exceeds AOA_{SS_TRIP} and the last calculated value of $V_{SS_UNFILTERED}$ equals or exceeds the current calibrated airspeed. The $V_{SS_UNFILTERED}$ value will remain latched until either the calibrated airspeed exceeds $V_{SS_UNFILTERED}$ or the angle of attack decreases below AOA_{SS_TRIP} . If either of these unlatching conditions occurs, a new value of $V_{SS_UNFILTERED}$ is calculated.

The top of amber band speed is set to the calculated Maneuver Margin above Stick Shaker value and is a function of $V_{SS_UNFILTERED}$, body-axis longitudinal acceleration, and body-axis normal acceleration. Calculated V_{SSNG} values are filtered prior to presentation to the flight crewmembers on the speed tape.

2.4 Maximum Gear and Flaps Extended Speeds

For barometric altitudes below 36,500 feet, the Maximum Gear Extended Speed (V_{LE}) is 270 knots. The Maximum Flaps Extended Speed (V_{FE}) schedule is shown in Table 1.

Table 1: Boeing 777-200ER Maximum Flaps Extended Speed, V_{FE}

Flap Position	Speed, knots
UP	255
5	235
15	215
20	195
25	185
30	170

2.5 Pilot Eye Height and Precision Approach Path Indicator (PAPI) Guidance

The pilot eye height parameter calculated by Boeing Engineering is based on airplane geometric altitude at the center of gravity with pitch and roll attitude adjustments to account for the pilot eye station. For calculation purposes, the pilot eye station is located 88.5 feet forward of the center of gravity and 3.33 feet above the center of gravity.

The Aircraft Performance Group Crash Site Factual Report documents the necessary data and calculations to determine the PAPI beam position relative to the runway 28L displaced threshold. For this study, the PAPI beam angle (2.98°) and intermediate deviation angles (average values of $\pm 0.185^\circ$) were based on the applicable July 2, 2013 FAA flight inspection report results. The PAPI maximum deviation angles ($\pm 0.583^\circ$) were based on the FAA guidance for Height Group 4 aircraft (Table 7-1 from AC 150/5340-30G). The longitudinal distance from the PAPI installation to the runway 28L displaced threshold is 1,366.5 feet.

3.0 RESULTS

3.1 Time Alignment

The time alignment between the FDR data and ADS-B data is summarized in Table 2 below. The Local Time value is based on the ADS-B data time code generated by a GPS clock.

Table 2: Time alignment mapping

Local Time	UTC Time	FDR Time
11:27:49.000	18:27:49.000	97911.657

The CVR to FDR time alignment solution was found to be linear over the approximately 10 minute length of the CVR recording selected. The linear solution required to satisfy the common event CVR/FDR constraints listed in Table 3 is presented in Table 4, where the time

Table 3: CVR/FDR common event constraints

CVR Time	FDR Low	FDR High	Event
01:49:57.664	97259.012	97260.012	begin transmission 14
01:50:02.507	97263.012	97264.012	end transmission 14
01:50:05.787	97266.012	97268.012	begin transmission 15
01:50:07.144	97268.012	97270.012	end transmission 15
01:50:15.035	97276.012	97277.012	begin transmission 16
01:50:20.883	97282.012	97283.012	end transmission 16
01:50:26.849	97287.012	97289.012	begin transmission 17
01:50:28.139	97289.012	97291.012	end transmission 17
01:52:29.493	97410.012	97411.012	begin transmission 18
01:52:32.940	97414.012	97415.012	end transmission 18
01:54:00.987	97502.012	97503.012	begin transmission 19
01:54:04.250	97505.012	97506.012	end transmission 19
01:54:56.170	97556.012	97558.012	begin transmission 20
01:54:57.066	97558.012	97560.012	end transmission 20
01:55:02.646	97563.012	97564.012	begin transmission 21
01:55:06.227	97567.012	97568.012	end transmission 21
01:56:23.608	97645.012	97646.012	begin transmission 22
01:56:28.105	97649.012	97650.012	end transmission 22
01:58:39.399	97780.012	97781.012	begin transmission 23
01:58:41.679	97783.012	97784.012	end transmission 23
01:58:55.986	97797.012	97798.012	begin transmission 24
01:59:00.833	97802.012	97803.012	end transmission 24
01:59:59.470	97860.012	97861.012	begin transmission 25
02:00:01.679	97862.012	97863.012	end transmission 25
02:00:10.722	97871.012	97873.012	begin transmission 26
02:00:12.402	97873.012	97875.012	end transmission 26

relationship is defined as

$$\text{CVR Time} = (\text{slope})(\text{FDR Time}) + \text{intercept.}$$

The uncertainty on the time alignment solution is less than ± 0.1 seconds. In Table 4, **CVR Time** in HH:MM:SS format is converted to total seconds (e.g., 01:56:44.276 = 7004.276 seconds).

Table 4: Solution to flight 214 CVR/FDR common event constraints

CVR Time		FDR Time, Seconds	Slope -	Intercept, Seconds
HH:MM:SS	Seconds			
01:56:44.276	7004.276	97665.680	1.00	-90661.404

3.2 Airplane Weight and Balance

The flight 214 FDR data indicated a ramp weight of 587,840 pounds, a takeoff weight of 586,560 pounds and a KSFO arrival weight of 423,360 pounds, for a fuel burn of about 163,200 pounds (24,358 gallons). The loading schedule indicated HL7742 weighed 586,703 pounds at takeoff of which 193,200 pounds (about 28,836 gallons) was fuel.

The Boeing Engineering assessment of flight 214 weight and balance in Figure 1 indicates an estimated time of arrival weight of about 421,200 pounds and a center of gravity position of 28.5 %MAC. The assumed fuel burn schedule maintained the airplane weight and center of gravity within the approved weight and balance envelope throughout the flight.

Based on the flight 214 load sheet, the center of gravity would have been located at 28.4 %MAC at takeoff. The FDR arrival weight of 423,360 pounds was used with the zero fuel weight to estimate that 29,857 pounds (4,456 gallons) of fuel remained at time of arrival. Adding the adjusted wing fuel weight (29,857 pounds) to the adjusted zero fuel weight (393,503 pounds) yields an arrival center of gravity location of 45.65 index units or 28.5 %MAC. Flight 214 calculated weight and center of gravity data are summarized in Table 5.

Table 5: Calculated flight 214 center of gravity locations

Flight 214 Load Sheet	Weight (LB)	ARM (Inches)	Index Units	Center of Gravity (%MAC)
Zero Fuel Wt.	393,503	1253.79	45.87	28.47
Takeoff Wt.	586,703	1253.62	43.60	28.41
Landing Wt.	421,204	1253.87	45.65	28.50

3.3 Speed Tape Parameter Calculations (Low Speed Amber Band)

The NTSB and Boeing calculated speed tape parameter results are compared in Figures 2 and 3. The calculated values for the filtered minimum maneuver speed (top of amber band speed), speed for EICAS caution alert AIRSPEED LOW,² and filtered stick shaker activation speed are shown for two time periods. The clean wing approach to airplane ground contact time segment is shown in Figure 2. The flaps 30, gear down time period is compared in Figure 3.

² If the airspeed drops below 30 percent of the difference between the filtered minimum maneuver speed and the filtered stick shaker activation speed, an EICAS caution alert is triggered which illuminates both MASTER CAUTION lights and generates an aural quadruple chime, an EICAS text message AIRSPEED LOW, and an airspeed value readout box color change (from white to amber) on the Primary Flight Display (PFD).

The Boeing and NTSB amber band parameter calculation results³ shown in Figures 2–3 are nearly identical. Both calculated minimum maneuver speed profiles compare favorably (within about 1 to 4 knots) with the sparse Flight Management Computer (FMC) minimum maneuver speed data recorded on the FDR. Differences between calculated (NTSB and Boeing) and recorded (flight 214 FDR) minimum maneuver speed parameters are likely due in part to the coarse resolution⁴ of the FDR angle of attack data (about $\pm 0.352^\circ$).

The sensitivity of the speed tape parameter calculations to the FDR combined left and right angle of attack resolution was estimated by adding a constant value of one-half the FDR angle of attack resolution (0.5 times 0.352) to the FDR-based angle of attack input signal at each time step and repeating the minimum maneuver speed calculation. The speed sensitivity to angle of attack results are plotted in Figures 4 and 5 with the added purple line identified as $V_{SSNG_dAOA=+0.175_deg_FILTERED_}(NTSB_CALC.)$. The addition of one-half the angle of attack parameter resolution yields a calculated minimum maneuver speed increase of about 1 to 4 knots for the available FDR comparison points.

Although the results are not shown in Figures 2 or 4, Boeing Engineering also reconstructed the $V_{SS_FILTERED}$ and $V_{SSNG_FILTERED}$ values for the earlier flaps-up portion of the descent (including the speed-brakes-up period). The Boeing and NTSB results reportedly differ by less than 1 knot, with the largest difference found during the speedbrake retraction transient.

In a separate study, Boeing Engineering used Warning Electronics System (WES) hardware, which calculates the $V_{SS_FILTERED}$ and $V_{SSNG_FILTERED}$ values onboard the airplane, in a laboratory bench test to validate the analytical calculation method outlined in Section 2.3. The Boeing analytical and hardware WES results matched well, with only small differences observed during dynamic conditions. Boeing Engineering attributed these small differences to filter implementation details and considered them to be insignificant for analytical method validation purposes.

In response to an NTSB request, Boeing Engineering provided a list of EICAS caution alerts that could result in a quadruple chime aural event on the accident airplane. The FDR data and the Boeing and NTSB speed tape parameter reconstruction results are consistent with an EICAS caution alert AIRSPEED LOW at FDR Subframe Reference Number (SRN) 97899.6. The CVR data indicate an aural quadruple chime event at FDR SRN 97900.7. A Boeing study found that the time delay between the calculated EICAS caution alert AIRSPEED LOW and the resulting aural quadruple chime on the aircraft should not exceed 1.8 seconds.

The FDR data and the Boeing and NTSB speed tape parameter reconstruction results are also consistent with stick shaker activation at FDR SRN 97907.6. The CVR data indicate that an aural stick shaker event began at FDR SRN 97907.8 and continued for at least 2.2 seconds. The calculated times for the Boeing and NTSB EICAS caution alert AIRSPEED LOW and expected stick shaker activation in this study are based on zero parameter resolution uncertainty in the applicable FDR parameters.

³ The Boeing low speed amber band calculated results are identified as $V_{SSNG_FILTERED}$, V_{LOW} , and $V_{SS_FILTERED}$ with respective black, red, and blue lines and matching color square, circular, and triangular symbols. The NTSB low speed amber band calculated results are identified as $V_{SSNG_FILTERED}$, V_{LOW} , and $V_{SS_FILTERED}$ with respective magenta, green, and orange lines and matching color square, circular, and triangular symbols.

⁴ The angle of attack resolution on the 256 words per second FDR installed on the accident airplane is 0.3515625 degrees.

3.4 Maximum Gear and Flaps Extended Speeds

A comparison of the maximum landing gear and flaps extended speed placards to the documented flight 214 landing gear and flap configurations is shown in Figures 6 and 7. The data indicate no landing gear or flap extension placard exceedances. However, the B777 Landing Attitude Modification (LAM) system reduced the scheduled flaperon droop in the landing flap configuration for a period of about 40 seconds (ending at about 35 seconds prior to ground impact, airspeed about 134 knots, altitude about 450 feet agl) while the aircraft was decelerating toward the AFM reference approach speed (V_{REF}).⁵

3.5 Pilot Eye Height and Precision Approach Path Indicator (PAPI) Guidance

A profile view of calculated pilot eye height relative to the calculated PAPI beam guidance is shown at the bottom of Figures 8–12 as a function of distance from the runway 28L displaced threshold (in nautical miles) and time prior to impact (in seconds). Each figure also includes FDR parameter time history data for airspeed, Mode Control Panel (MCP) selected speed, autopilot engaged/disengaged status, left and right flight director on/off status, pitch mode, autothrottle mode, left and right engine throttle lever angle, left and right engine N1, and left and right stick shaker. In addition, the FDR sink rate and MCP selected vertical speed parameters are plotted in Figures 8–11.

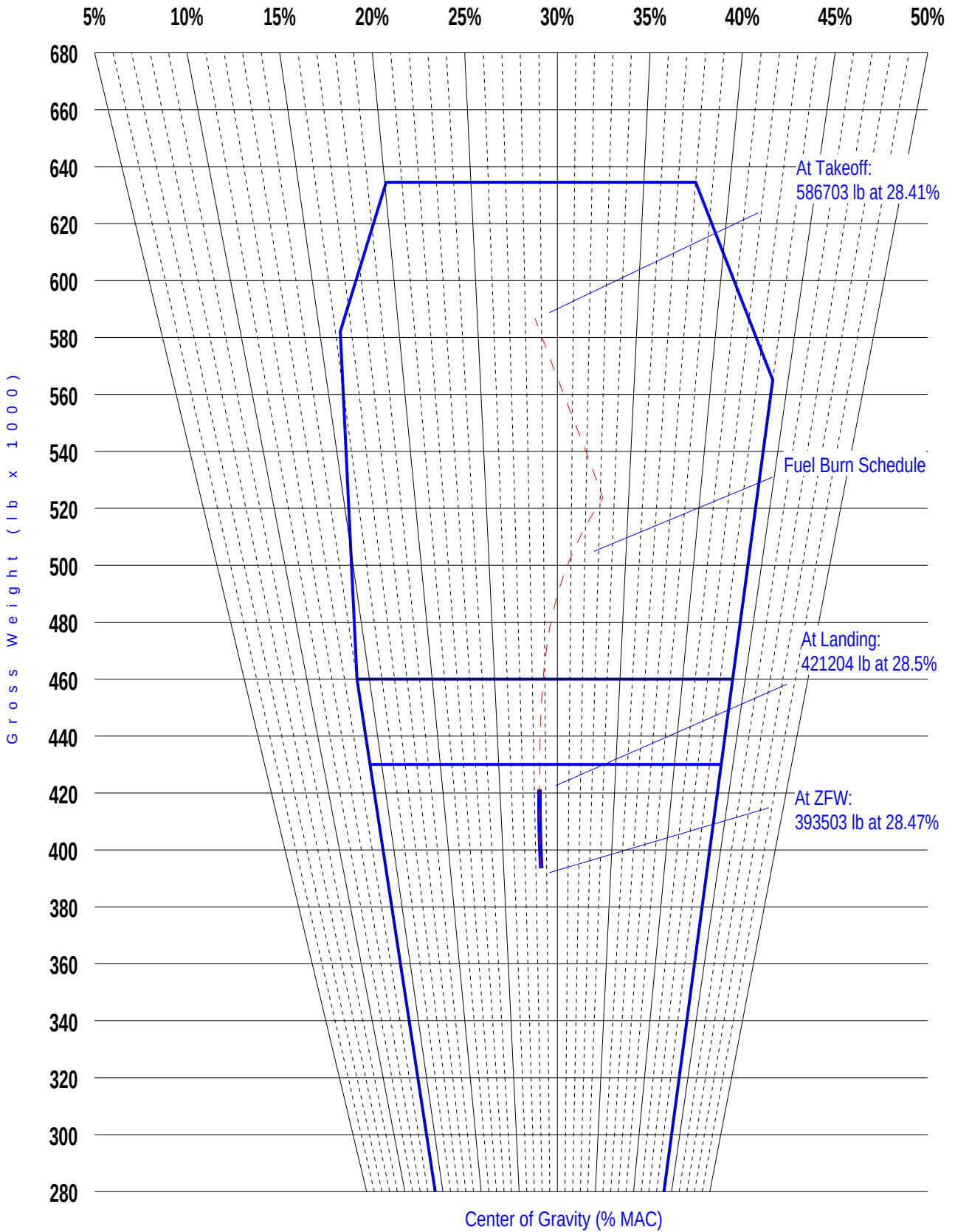
Flight path and FDR parameters are presented in Figures 8–10 as a function of Asiana flight 214 remaining distance and time using altitude scale limits of 2,500; 1,000; and 600 feet; respectively. The FDR pitch attitude and flight director incremental pitch guidance parameters are shown in Figures 11–12 using the 600-foot altitude scale limit format. Recorded control column position (6 parameters) and left and right control column force (2 parameters) are provided in Figure 12.

4.0 SUMMARY

The event flight time alignment between the CVR, FDR, ATC communication recordings, ADS-B, and ASR radar data was determined. Based on the load sheet component weight and balance buildup and the FDR weight parameters, the center of gravity at time of arrival was located at about 28.5 %MAC, near the middle of the envelope. A comparison of the maximum landing gear and flaps extended speed placards to the documented flight 214 landing gear and flap configurations indicates no landing gear or flap extension placard exceedances.

The NTSB and the Boeing calculated minimum maneuver speed profiles compare favorably (within about 1 to 4 knots) with the sparse Flight Management Computer (FMC) minimum maneuver speed data recorded on the FDR. In addition, the calculated timing results for the expected EICAS caution alert AIRSPEED LOW and the stick shaker activation events appear to match the available CVR data within expected timing tolerances for each event. Calculated pilot eye height, calculated PAPI guidance, and select FDR parameters were plotted to visualize the airplane flight path and energy state as a function of Asiana flight 214 distance from the runway 28L displaced threshold and time prior to impact.

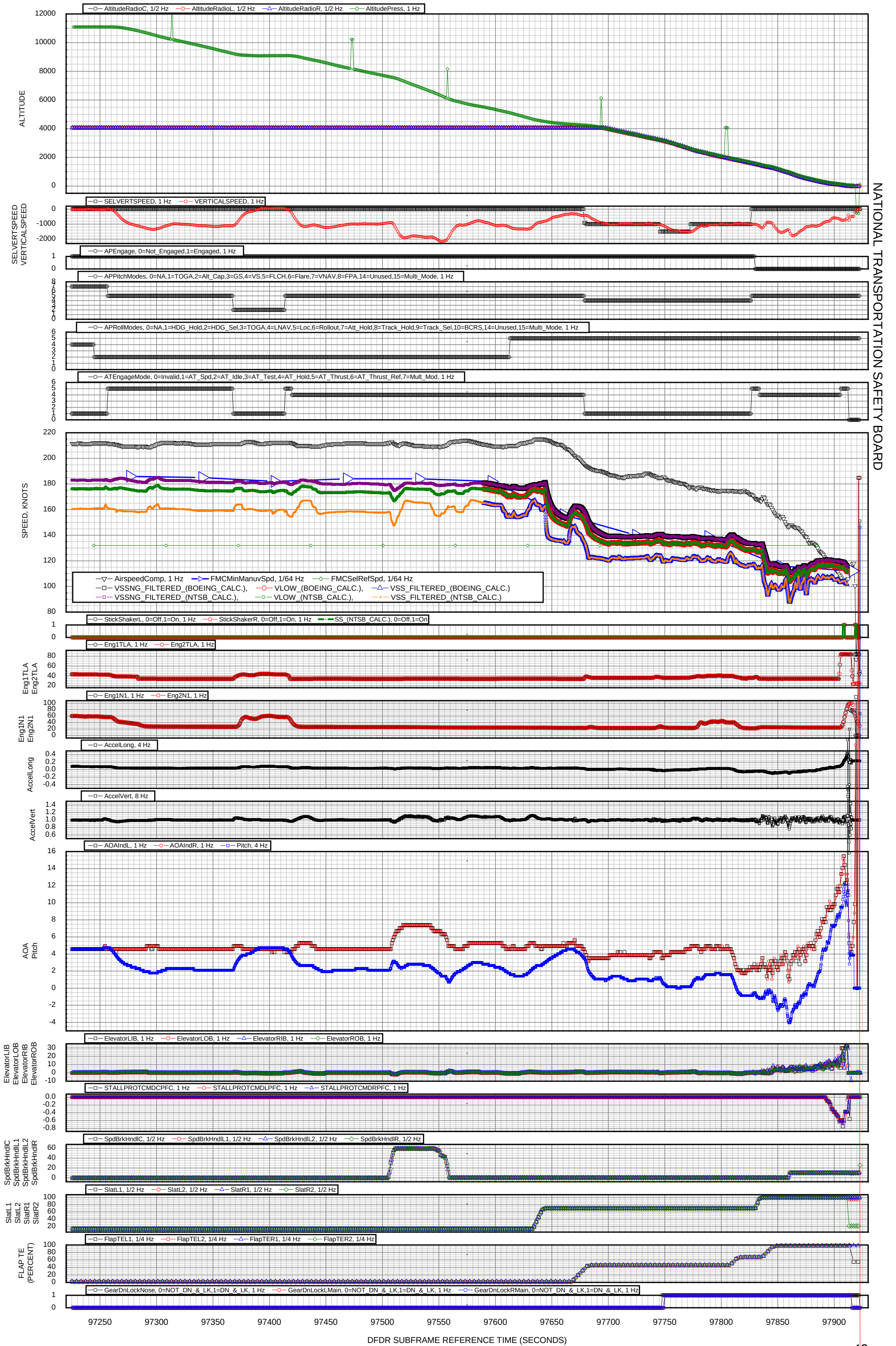
⁵ The LAM system is designed to reduce the likelihood of nose-first contact under certain speed and weight conditions. Landing attitude modification begins at or slightly faster than V_{REF} and is fully in place at speeds 20 knots above V_{REF} . A portion of the approach is commonly flown at $V_{REF}+20$ as the airplane decelerates toward the approach speed or, for example, with autothrottle off and required wind additives.



CALC	G W Gee	29 AUG 13	REVISED	DATE	ASIANA INCIDENT AT SAN FRANCISCO	777-200ER
CHECK						FIGURE 1
APPD.						PAGE
APPD.						11

BOEING

FIGURE 2: ASIANA AIRLINES BOEING 777-200ER IMPACT WITH SEA WALL AND RUNWAY 28L ON FINAL APPROACH
SAN FRANCISCO INTERNATIONAL AIRPORT (KSFO), SAN FRANCISCO, CA; JULY 6, 2013 [SPEED TAPE CALCULATIONS]



NATIONAL TRANSPORTATION SAFETY BOARD

FIGURE 3: ASIANA AIRLINES BOEING 777-200ER IMPACT WITH SEA WALL AND RUNWAY 28L ON FINAL APPROACH
SAN FRANCISCO INTERNATIONAL AIRPORT (KSFO), SAN FRANCISCO, CA; JULY 6, 2013 [SPEED TAPE CALCULATIONS]

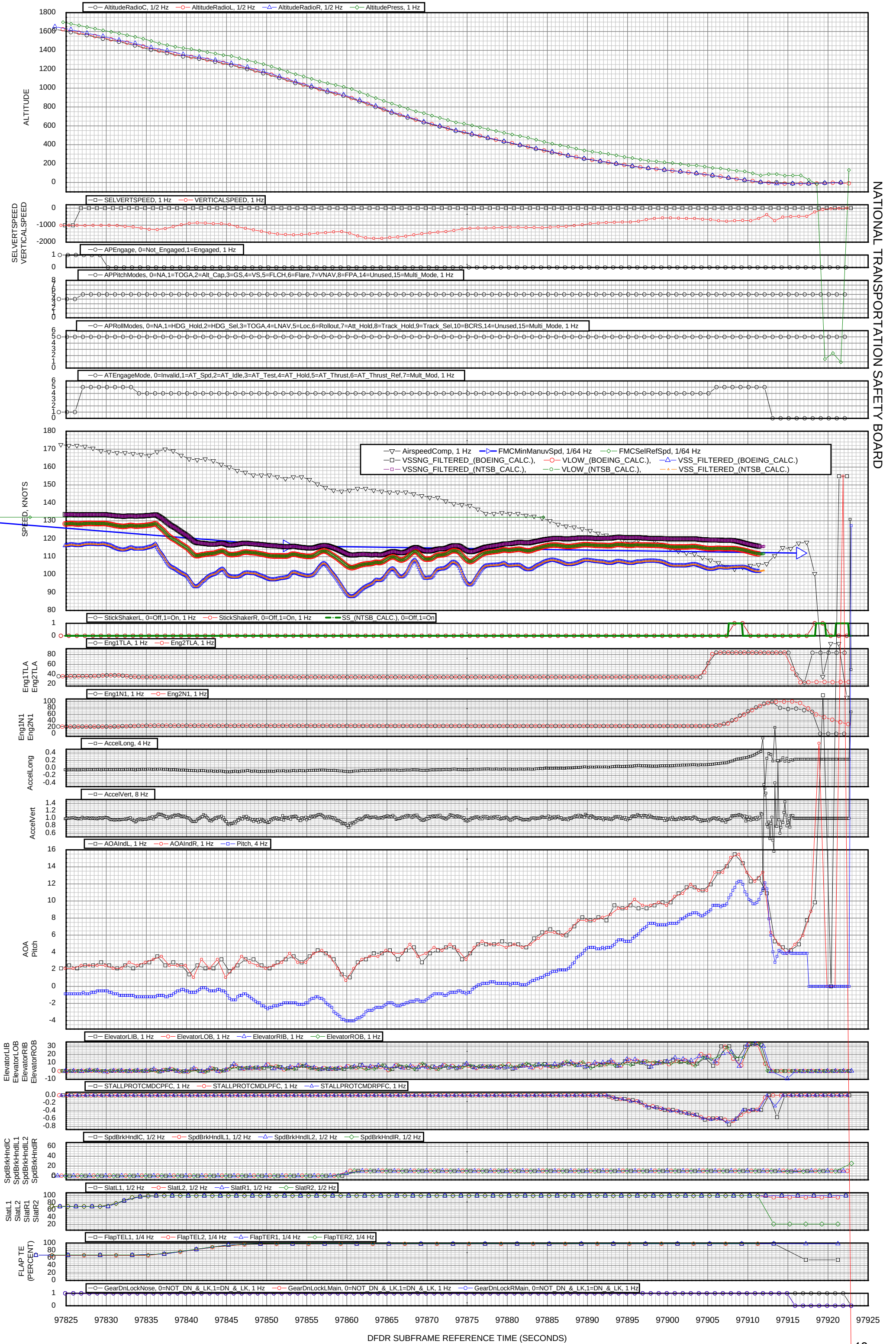


FIGURE 4: ASIANA AIRLINES BOEING 777-200ER IMPACT WITH SEA WALL AND RUNWAY 28L ON FINAL APPROACH
SAN FRANCISCO INTERNATIONAL AIRPORT (KSFO), SAN FRANCISCO, CA; JULY 6, 2013 [SPEED TAPE CALCULATIONS, ANGLE OF ATTACK SENSITIVITY]

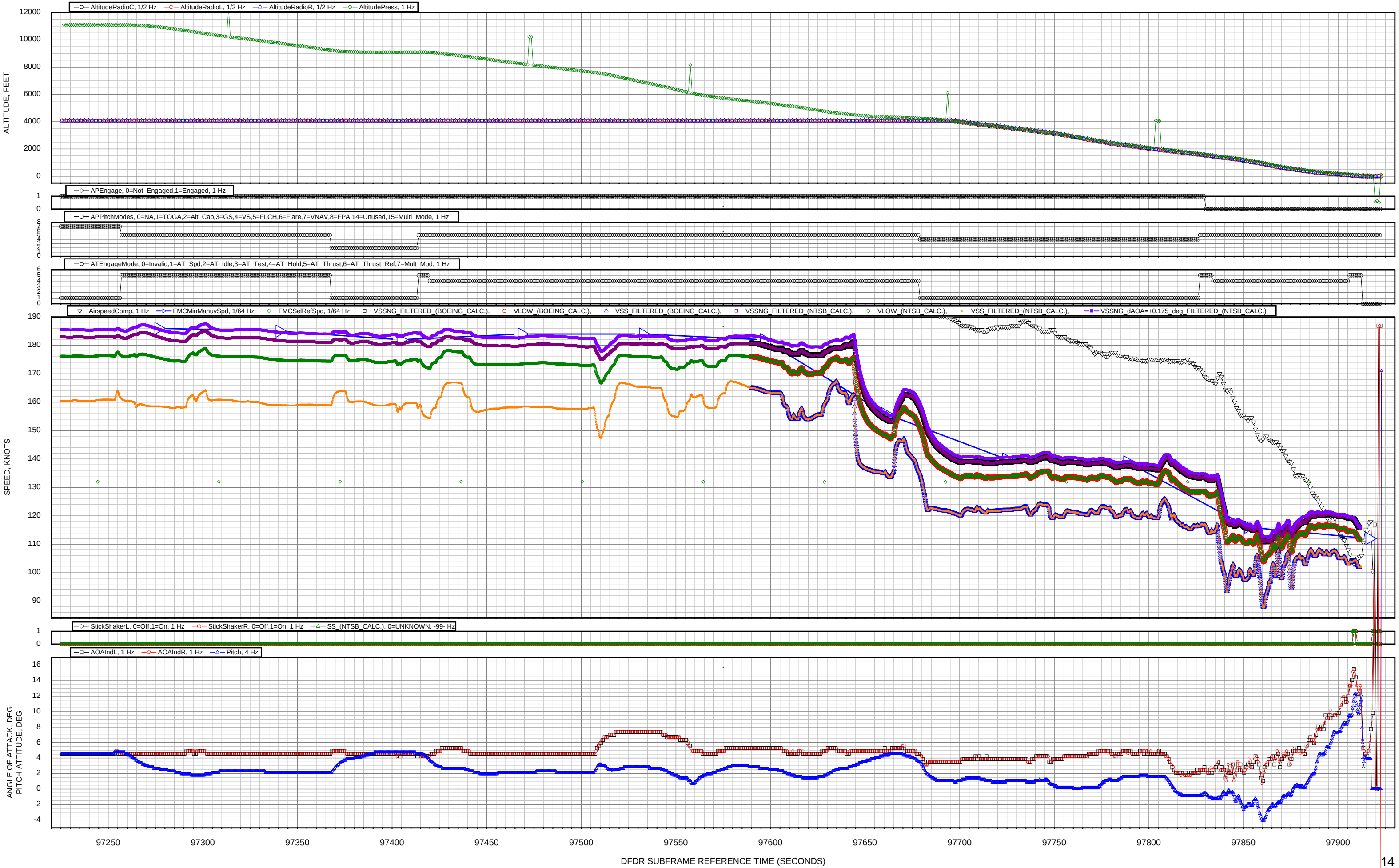


FIGURE 5: ASIANA AIRLINES BOEING 777-200ER IMPACT WITH SEA WALL AND RUNWAY 28L ON FINAL APPROACH
SAN FRANCISCO INTERNATIONAL AIRPORT (KSFO), SAN FRANCISCO, CA; JULY 6, 2013 [SPEED TAPE CALCULATIONS, ANGLE OF ATTACK SENSITIVITY]

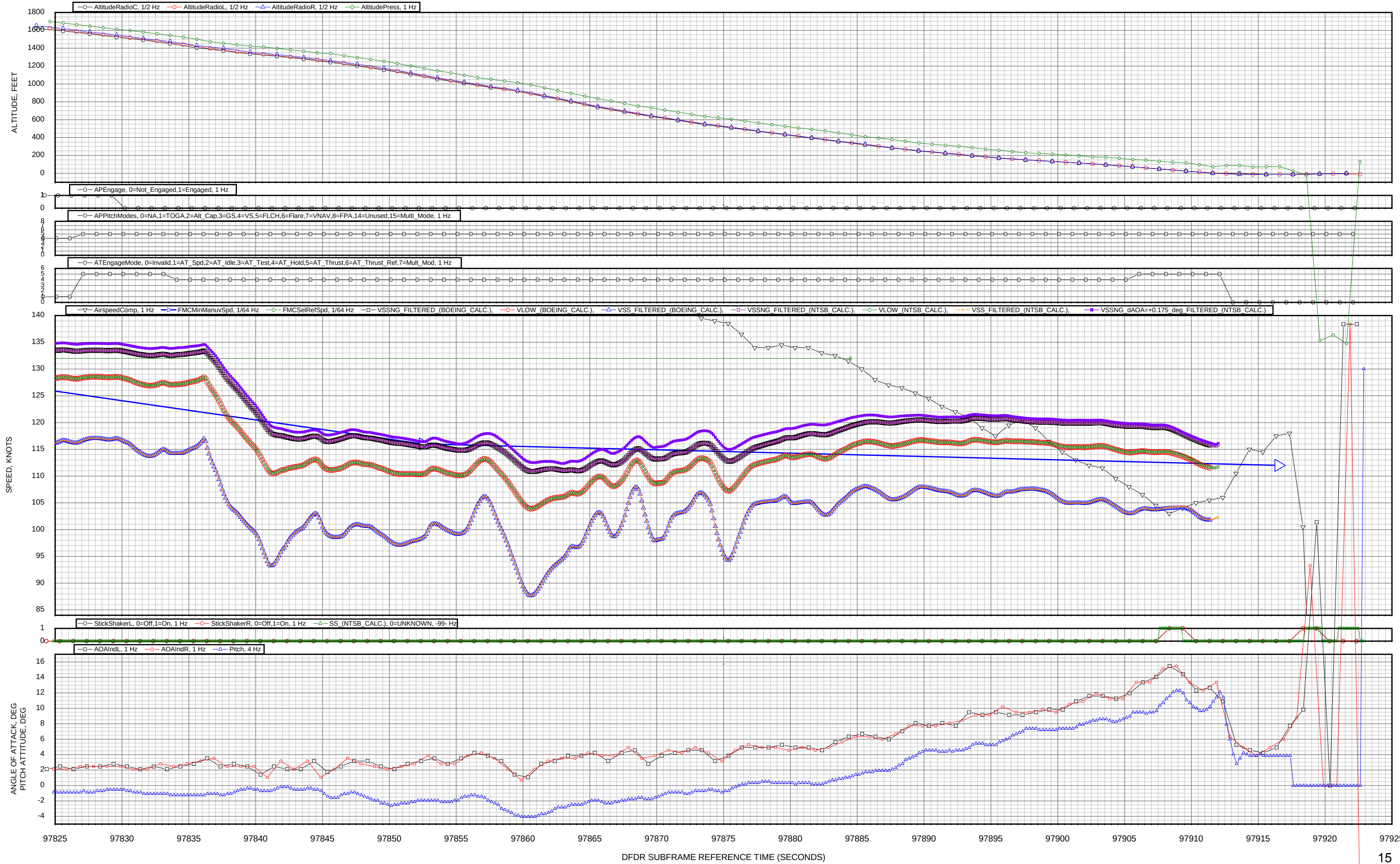
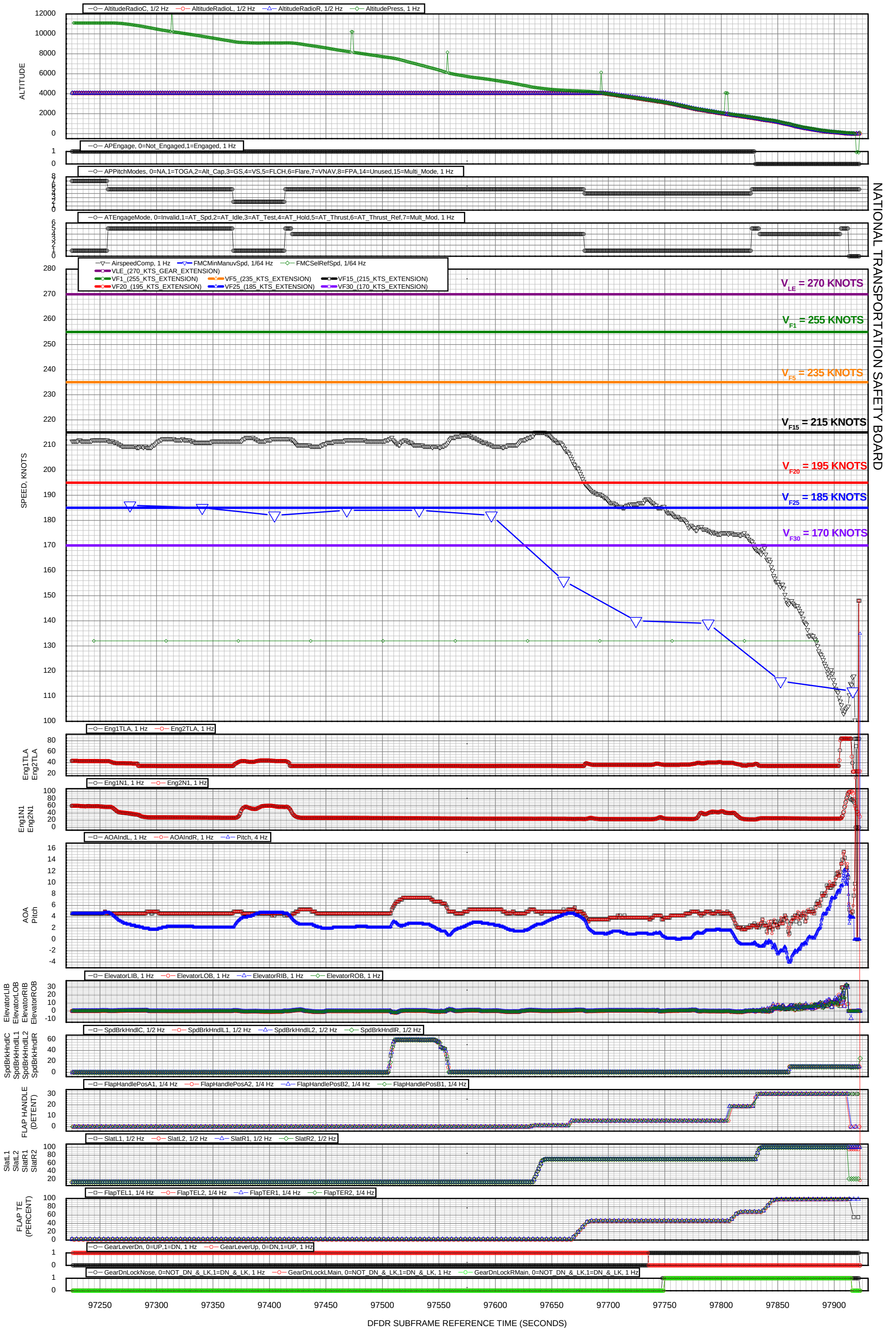
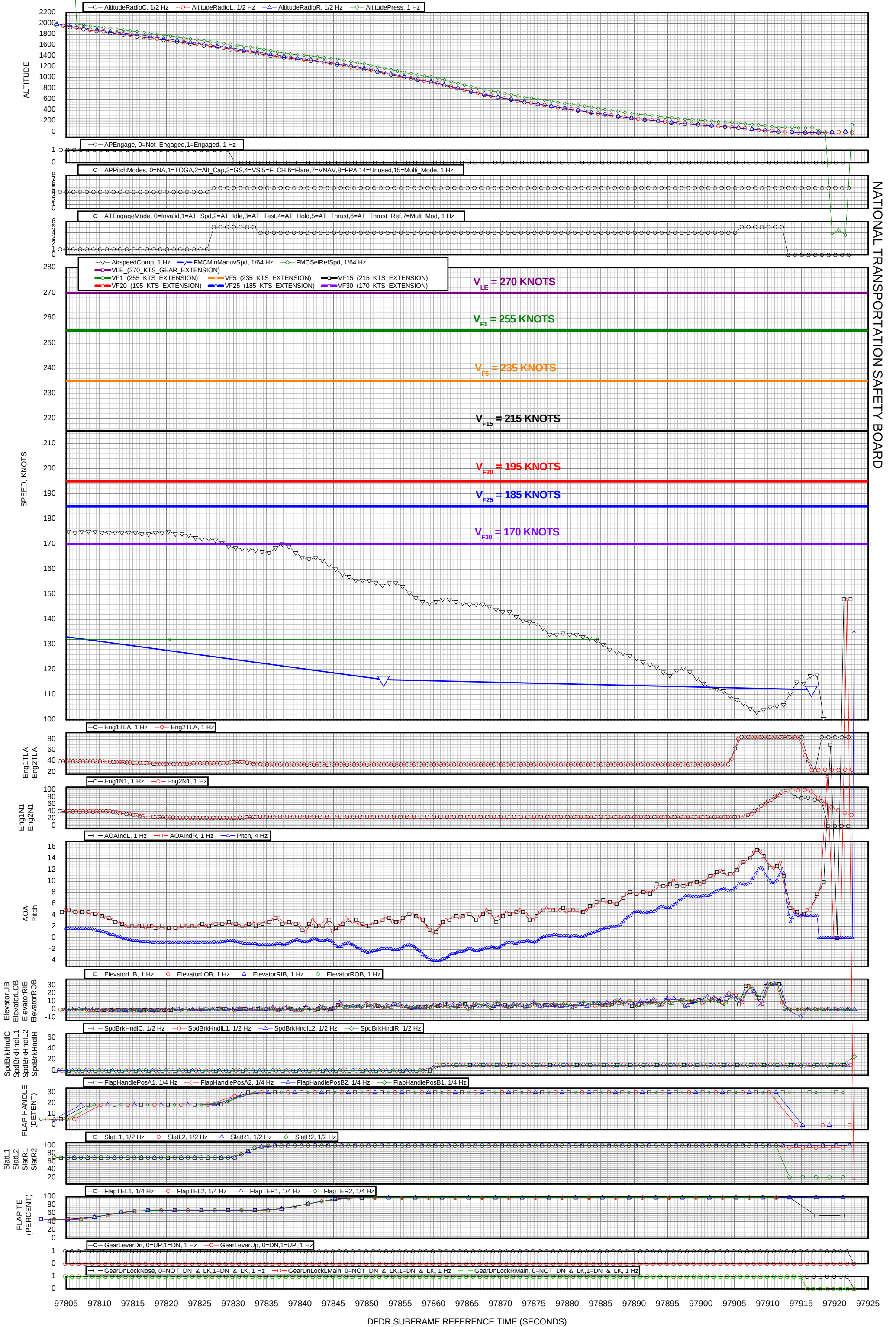


FIGURE 6: ASIANA AIRLINES BOEING 777-200ER IMPACT WITH SEA WALL AND RUNWAY 28L ON FINAL APPROACH
SAN FRANCISCO INTERNATIONAL AIRPORT (KSFO), SAN FRANCISCO, CA; JULY 6, 2013 [LANDING GEAR AND FLAP EXTENSION PLACARDS]

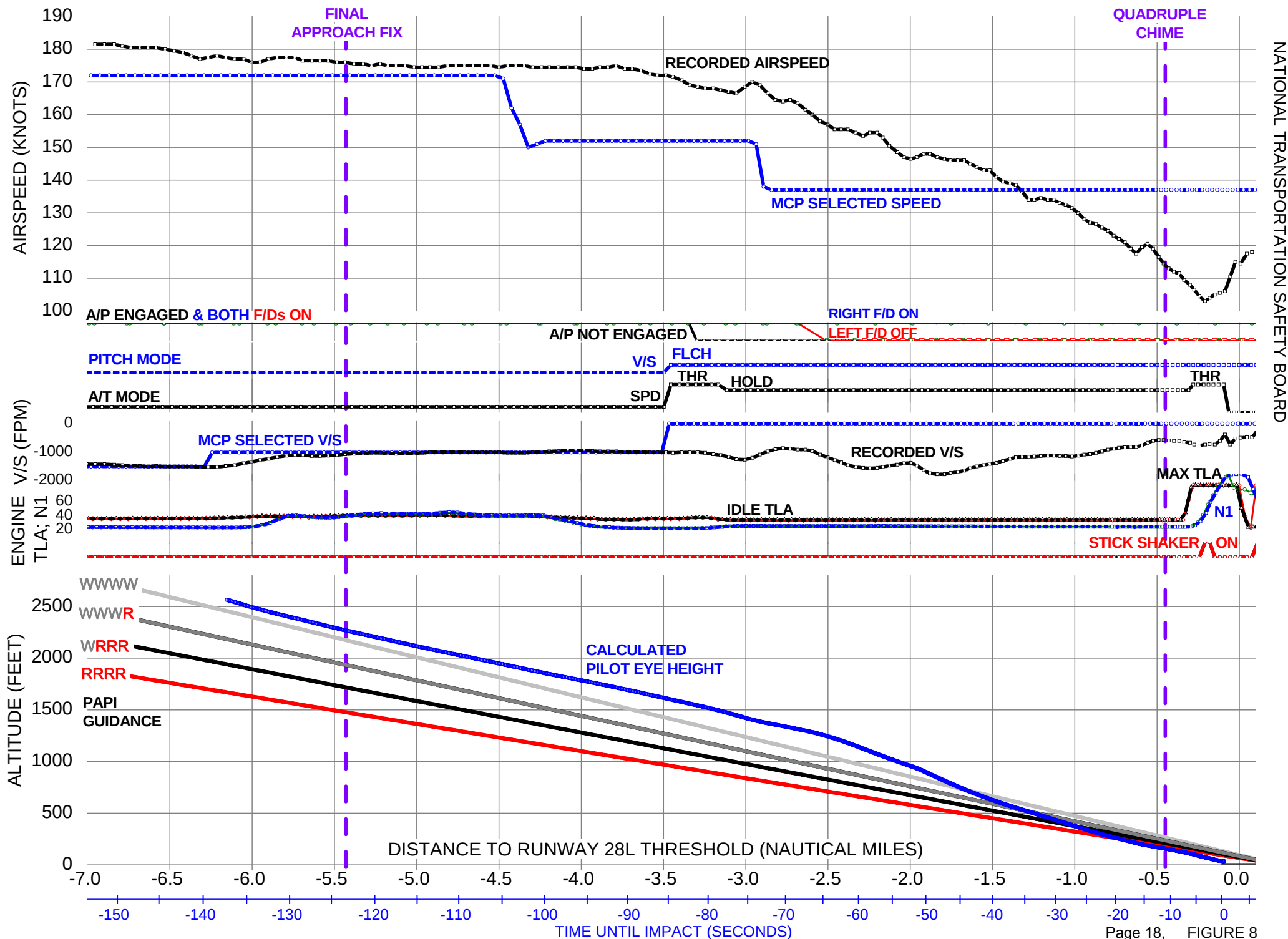


NATIONAL TRANSPORTATION SAFETY BOARD

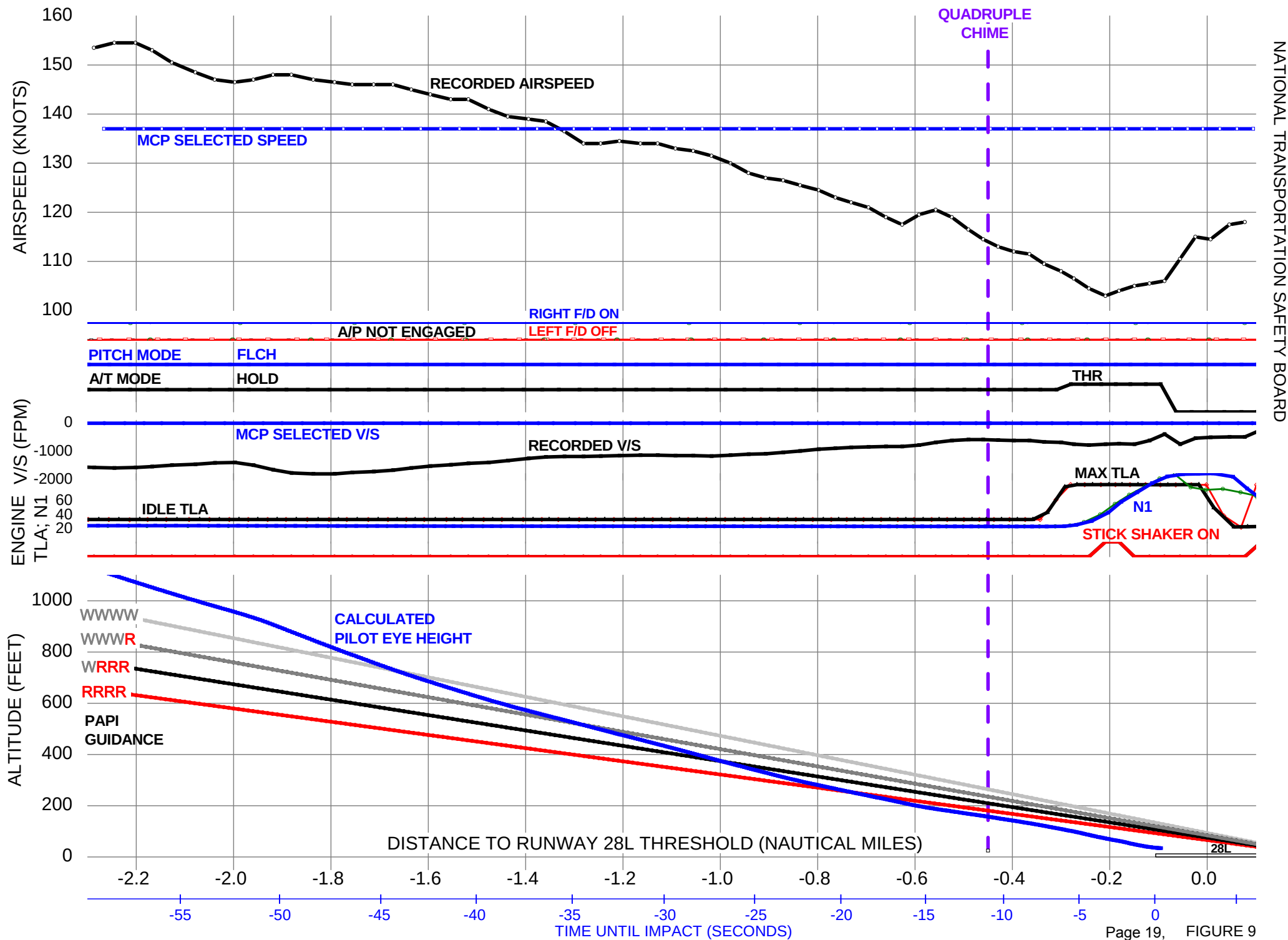
FIGURE 7: ASIANA AIRLINES BOEING 777-200ER IMPACT WITH SEA WALL AND RUNWAY 28L ON FINAL APPROACH
SAN FRANCISCO INTERNATIONAL AIRPORT (KSFO), SAN FRANCISCO, CA; JULY 6, 2013 [LANDING GEAR AND FLAP EXTENSION PLACARDS]



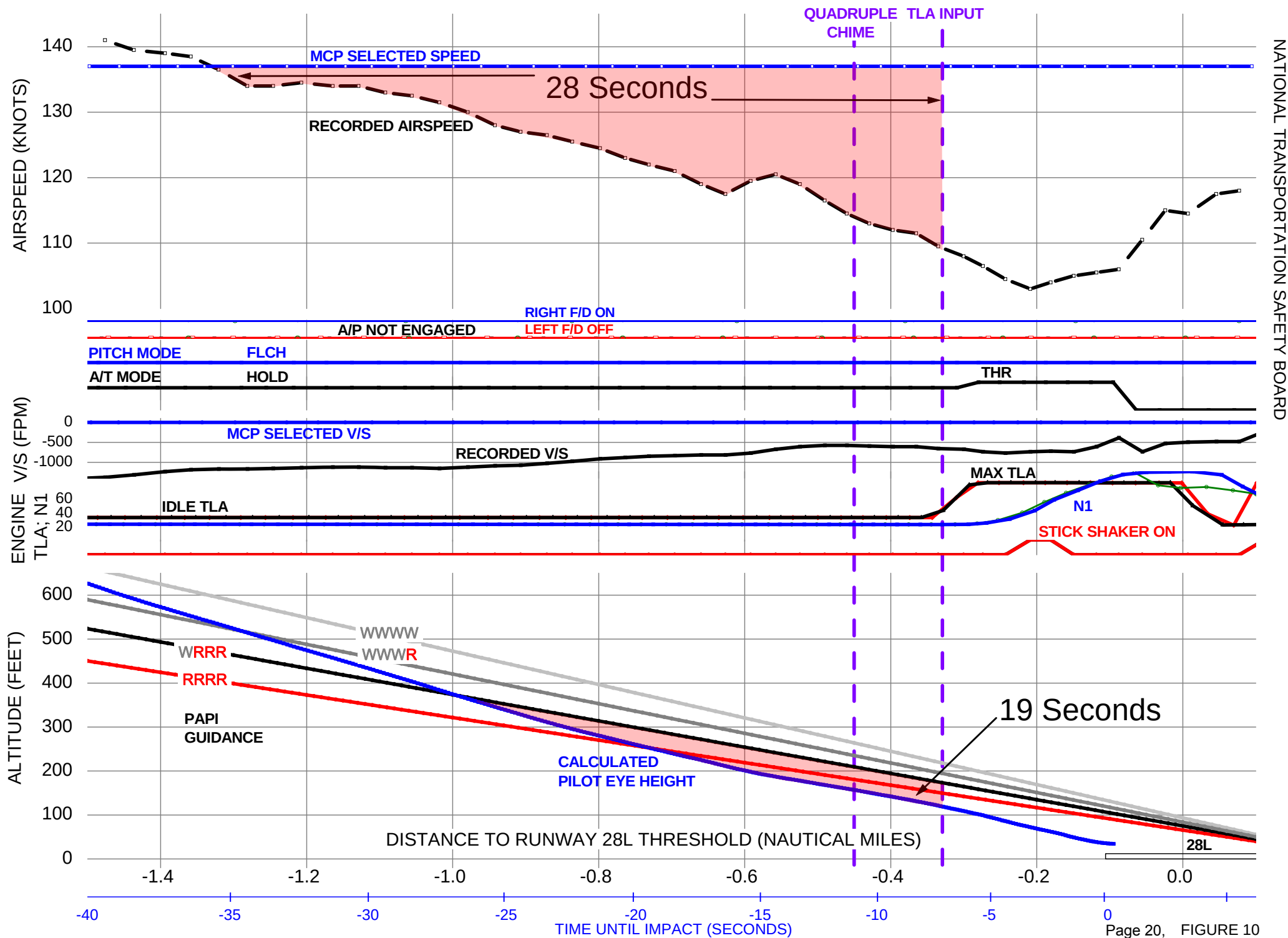
ASIANA AIRLINES BOEING 777-200ER FLIGHT 214 FINAL APPROACH; JULY 6, 2013



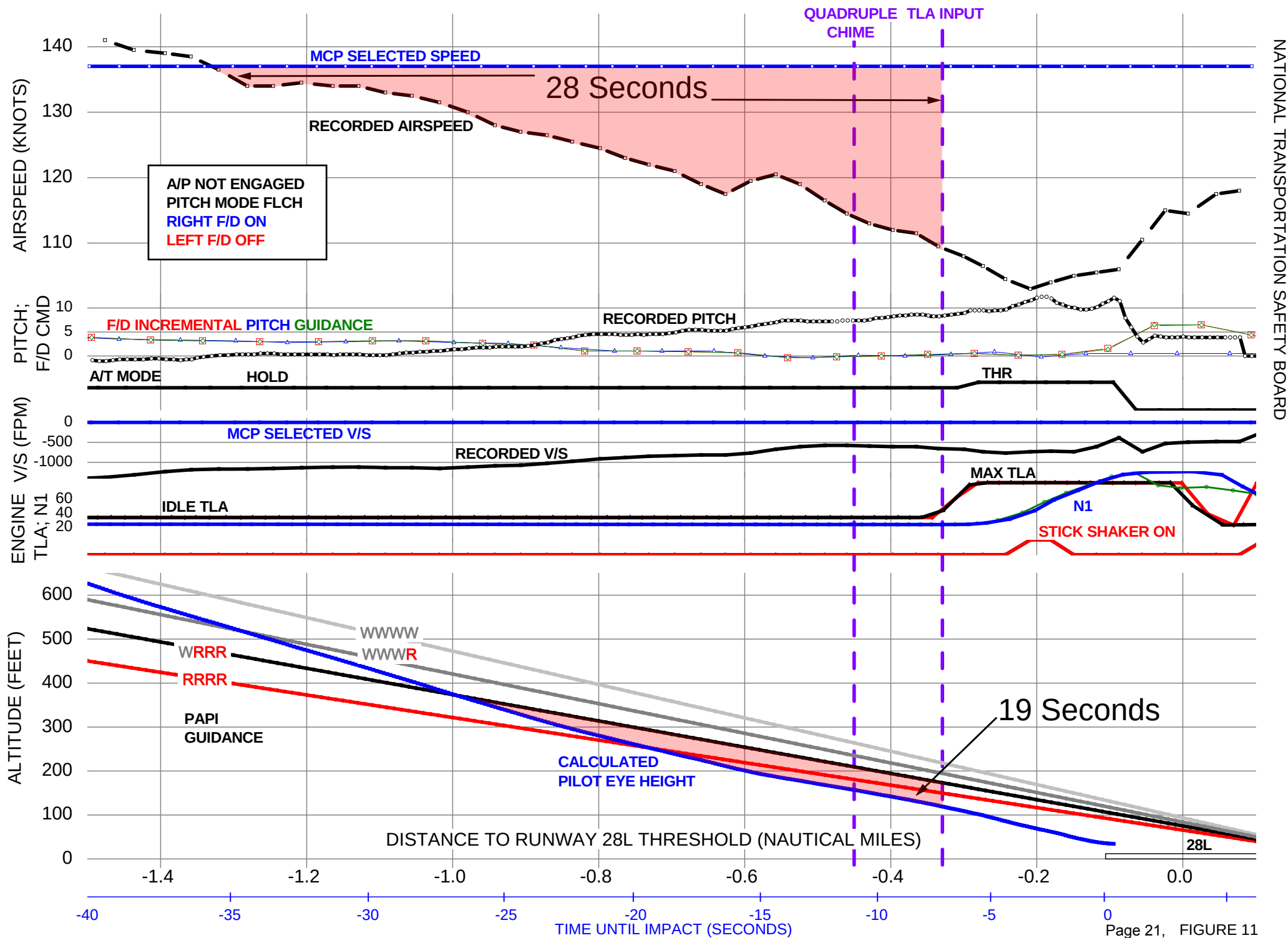
ASIANA AIRLINES BOEING 777-200ER FLIGHT 214 FINAL APPROACH; JULY 6, 2013



ASIANA AIRLINES BOEING 777-200ER FLIGHT 214 FINAL APPROACH; JULY 6, 2013



ASIANA AIRLINES BOEING 777-200ER FLIGHT 214 FINAL APPROACH; JULY 6, 2013



ASIANA AIRLINES BOEING 777-200ER FLIGHT 214 FINAL APPROACH; JULY 6, 2013

